BAEC ENGINE 2

Monitoring + Correspondence

Research and Technical Brief

Project: BAEC Trigger Intelligence

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Software implementation accompanying the conceptual working paper

When the Buyer Is Not In-Market: Buyer-Articulated Evaluation Contingencies in B2B Sales Encounters

Remember what the buyer said mattered, then watch for it.

Abstract

Business-to-business sales encounters often occur before a buyer has decided that evaluating alternatives is worthwhile. The accompanying conceptual working paper defines a buyer-articulated evaluation contingency (BAEC) as information communicated by a presently non-evaluating organizational buyer about a prospective condition that would make initiating or reopening alternative evaluation worthwhile. BAEC Engine 2 is a research software prototype that begins after Engine 1 has confirmed such a buyer-defined condition. It operationalizes human-authorized monitoring of legitimate evidence sources, preservation of corrections and provenance, assembly of possible correspondence, and human review followed by deterministic classification. The public demonstration uses a fixed synthetic scenario, makes no live model request, uses no real customer data, and stops before any seller action. This brief documents the theoretical basis, authority boundaries, software architecture, worked synthetic case, verification status, limitations, and a reproducible click-by-click demonstration protocol. Software verification is evidence about implementation behavior only. It does not validate the BAEC construct, establish purchase intent, or show that correspondence creates an opportunity.

Source hierarchy and claim status

The conceptual working paper is the theoretical source of truth. The Research Contract translates the paper into software-facing constraints; if the two conflict, the working paper prevails. Engine 2 implementation behavior and verification artifacts demonstrate software behavior, not empirical validity. This brief uses the same claim-status convention as the Engine 1 brief so that theory, managerial interpretation, and software design are not presented as equivalent forms of evidence.

Table 1. Claim-status convention used in this brief.

Status	Meaning	Examples in this brief
DEFINITIONAL	Part of the BAEC or CEE construct definition.	BAEC definition; evaluation activation endpoint.
PROPOSED	A theoretical proposition or future extension requiring empirical testing.	Prospective validity; AI-enabled monitoring tests.
MANAGERIAL	An account-management interpretation proposed from BAEC information.	Preserve the condition; selective reassessment; buyer-specific monitoring.
IMPLEMENTATION	A software choice made to preserve evidence, authority, or auditability.	Monitoring Plan; correspondence dimensions; append-only stores; locked classifier.

1. Research problem and conceptual basis

Many B2B sales interactions occur before a buyer has entered active search or comparison. Once a buyer-defined contingency has been captured and confirmed, a later account-management question emerges: how should new evidence be assessed when it may correspond to the condition the buyer previously identified? Engine 2 narrows this problem to buyer-specific monitoring and human reassessment rather than generic intent prediction.

Engine 2 research question

> *Has new evidence appeared that may correspond to a verified buyer-defined condition, and can that correspondence be reviewed without converting it into purchase intent, an opportunity, or permission to act?*

Two engines, two questions

Table 2. Separation between Engine 1 and Engine 2.

Engine	Primary question
Engine 1	What did the buyer say would make evaluation worthwhile, and can that condition be verified and confirmed?
Engine 2	Has new evidence appeared that may correspond to that verified buyer-defined condition?

Critical boundary

> *Correspondence narrows human attention. It does not establish that the buyer is currently evaluating, intends to purchase, or that an opportunity exists.*

The distinctive idea is not the existence of monitoring. It is the origin of the signal specification: the buyer's prior statement rather than only a seller-defined trigger or a model-inferred pattern. Engine 2 is therefore designed to preserve source, threshold, timing, provenance, and human authority while stopping before seller action.

2. Engine 1 to Engine 2 handoff

Engine 2 begins only after Engine 1 has persisted a CONFIRMED_BAEC. The Stage K handoff is read-only, point in time, and deliberately narrow. Engine 2 does not re-run confirmation and does not write back to Engine 1.

Table 3. Handoff rules.

Boundary	Rule
Eligibility	A persisted CONFIRMED_BAEC with its human confirmation is required. AI drafts, accepted-only reviews, and unexecuted grants do not cross.
Buyer wording	The buyer's exact statement is the Engine 2 condition reference; a paraphrase does not replace it.
Threshold precision	Comparator, numeric value, unit, and stated timing are preserved. A numeric value without a unit is refused rather than guessed.
Currentness	Engine 1 remains authoritative. Monitoring activates only when the supplied staleness status is CURRENT.
Synchronization	The handoff is a snapshot. Later Engine 1 changes are seen only through a new read.
Mutation	Engine 2 writes nothing to Engine 1, consumes no grant, and creates no account-state transition.

A confirmed BAEC is not an opportunity. A HUMAN_VERIFIED_CORRESPONDENCE result is also not an opportunity. Both boundaries are explicit and tested.

3. Core architecture and authority

Engine 2 separates monitoring configuration, raw evidence, arithmetic, candidate assembly, human judgment, deterministic outcome, and optional AI advice. Each layer carries only the authority assigned to it.

Table 4. Engine 2 concepts and limits.

Concept	What it is	What it is not
Confirmed BAEC handoff	Read-only snapshot of one confirmed Engine 1 BAEC.	A new BAEC, re-confirmation, or live synchronization.
Monitoring Plan	Human-authorized rules, required dimensions, sources, threshold, and timing.	Evidence that the condition occurred.
Authorized Source	A source approved for one plan with basis, scope, and limitations.	Permission to contact anyone.
Observation	Exact captured source text with provenance and integrity digest.	An interpretation or editable note.
Supersession / retraction	A later record correcting or withdrawing an earlier Observation while preserving history.	Deletion.
Derived Measurement	Deterministic arithmetic from source Observations.	A free-form model calculation.
Signal Candidate	Evidence selected as possibly relevant.	A finding.
Correspondence Candidate	A hypothesis that evidence may bear on the BAEC and warrants human review.	A lead, opportunity, or intent signal.
Human Correspondence Review	A person's findings, quality checks, sufficiency judgment, and revalidation flag.	An account-state change or seller action.
Correspondence outcome	A locked deterministic result derived from the human review.	A label a caller can choose.
Advisory AI	An optional validated proposal stored separately.	A human finding or authoritative outcome.

Evidence lifecycle

Confirmed buyer condition → human-authorized Monitoring Plan → Authorized Sources → immutable Observations → preserved corrections, retractions, and refusals → usable evidence ledger → Signal Candidate → Correspondence Candidate → Human Correspondence Review → deterministic correspondence outcome.

Authority model

Engine 1 is authoritative for the buyer evidence and confirmed BAEC. Engine 2 capture is authoritative only that exact text was captured under an active authorized plan. Signal and Correspondence Candidates assert possible relevance only. The

human reviewer is authoritative for the human findings. The locked classifier is authoritative only for the narrow correspondence outcome derived from those findings.

4. Human review and locked outcome logic

The public demo asks the visitor to make every human judgment required by the locked review model. Nothing is pre-filled, no AI suggestion is used, and the caller does not submit an outcome. The review covers 11 correspondence dimensions, four cross-cutting quality checks, evidence sufficiency, and whether the buyer condition should be revalidated.

Table 5. Correspondence dimensions.

Dimension	Question
Entity match	Is the evidence about the entity the condition concerns?
Relationship match	Does it concern the buyer's own relationship or agreement?
Source relevance	Is the source capable of establishing the relevant fact for this buyer?
Event or condition type	Is the reported event the kind of change the buyer named?
Threshold match	Does the reported magnitude meet the buyer's threshold?
Comparator match	Is the buyer's comparator applied exactly, including strict "more than"?
Numeric magnitude	Is the magnitude established exactly as stated?
Unit match	Is the evidence in the buyer's unit?
Timing match	Does the evidence satisfy the buyer's timing context?
Renewal or effective-date match	Does the effective date align with the stated renewal or other time?
Causal or contextual relevance	Is the match substantive rather than coincidental?

Cross-cutting checks

- Freshness: whether the evidence is current rather than superseded, retracted, or outdated.
- Contradictory evidence: whether accepted evidence contradicts the proposed correspondence.
- Missing evidence: whether anything required to resolve the review is missing.
- Provenance quality: whether evidence comes from an authorized source with complete provenance.

Locked outcomes

Table 6. Deterministic correspondence outcomes.

Outcome	Rule
NO_CORRESPONDENCE	At least one required dimension is found contradicted.
POSSIBLE_CORRESPONDENCE	A required dimension remains unresolved, evidence is not judged sufficient, or BAEC revalidation is required.
HUMAN_VERIFIED_CORRESPONDENCE	Every required dimension is supported, evidence is judged sufficient, and BAEC revalidation is not required.

There is no override field, probability, confidence score, or "likely" outcome. A domain-invalid review is refused rather than converted into a fourth correspondence outcome.

5. Public synthetic demonstration

The public demo uses an original synthetic account and supplier. It is not drawn from the research corpus and should not be read as evidence about real buyer behavior.

Buyer-defined condition

> *"If Cobalt raises the price of our imaging service contract by more than 8% at the July renewal, we would put the contract out for review."*

Synthetic account: Alder Ridge Clinics. Synthetic supplier: Cobalt Imaging Supply. Threshold: more than 8%. Timing: July renewal.

Authorized sources

Table 7. Authorized sources in the demonstration.

Source	Authorization basis	Limitation
Supplier renewal notice	Forwarded by the buyer under the account agreement.	Contract-specific notice text and dates.
Supplier public announcement	Public announcements on the supplier's news page.	May describe all customers rather than this contract.

Evidence history

Table 8. Synthetic evidence and usability.

ID	Evidence summary	Status
OBS-DEMO-04	12 Apr: supplier says existing-customer service prices will remain flat through 2027.	RETRACTED
OBS-DEMO-05	14 Apr: supplier retracts the 12 Apr price-freeze statement.	CURRENT / USABLE
OBS-DEMO-03	20 Apr: supplier announces price adjustments up to 10% for second-half 2027 renewals.	CURRENT / USABLE context
OBS-DEMO-01	3 May renewal notice: USD 40,000 → USD 44,800, yielding 12.00%.	SUPERSEDED; measurement invalidated
OBS-DEMO-02	10 May corrected renewal notice: USD 40,000 → USD 43,600 effective 1 Jul 2027, yielding 9.00%.	CURRENT / USABLE
ITEM-DEMO-FORUM-01	Forum rumor that Cobalt is raising prices 15% for everyone.	REFUSED; unauthorized source

The critical public-demo sequence is 12.00% initially reported → corrected → 9.00% current. The 12.00% record remains visible in history but is not usable after the correction.

6. Advisory AI boundary

Engine 2 contains an optional advisory-AI architecture, but AI is structurally separated from authoritative human review. The public demo does not call a model and does not require an AI-provider credential.

- Stage F builds a curated canonical input from the buyer's exact statement, Monitoring Plan, and evidence.
- AI output is limited to a proposal vocabulary: PROPOSED_SUPPORTED, PROPOSED_CONTRADICTED, PROPOSED_UNRESOLVED, and PROPOSED_NOT_APPLICABLE.
- An AI proposal has no outcome, no evidence-sufficiency judgment, and no authoritative human finding.
- Stage G stores AI inputs, attempts, exact responses, validation, and accepted proposals in the separate BAEC_ENGINE_2_AI store.
- Stage I may display an accepted proposal beside the human workflow only on explicit request; nothing flows into the human fields.
- Stage H-A provides an opt-in controlled live harness. Stage H-B has not been executed.

The public demonstration therefore uses no live AI call, no external database, no customer upload, no CRM integration, and no seller-action control.

7. Verification, reproducibility, and limitations

Verification snapshot

Table 9. Current software verification state.

Item	Value
Engine 2 v1.0 release tag	baec-engine2-v1.0 → b820778108f7c9432689755c4b255f9082d24350
Public demo commit	df77f38cbb64729b5e1579af136431d448dd9bc3
Engine 1 schema	version 7
BAEC_ENGINE_2	schema version 1; 30 tables
BAEC_ENGINE_2_AI	schema version 1; 7 tables
Public-demo tests	34 passed
All Engine 2 tests	979 passed
Full repository suite	6,797 passed; 2 opt-in live tests deselected; 0 skipped

Synthetic corpus distribution	20 NO; 32 POSSIBLE; 9 HUMAN_VERIFIED; 3 blocked
Engine 1 dev database SHA-256	add117f8452adbcf96368ae5fd57bc71be3b83493fb4136d96aea84834e1ef06
Stage H-B	Paused; not executed

Reproduction commands

python3 scripts/engine2_release_demo.py — runs the deterministic Engine 1 → Engine 2 release demonstration in memory.

python3 scripts/verify_engine2_release.py — verifies identities, integrity, schemas, demonstration, and focused suites.

python3 scripts/verify_engine2_release.py --full — adds the full repository suite.

streamlit run streamlit_engine2_demo.py — runs the public LinkedIn-facing demonstration locally.

Key limitations

- The conceptual working paper is unpublished and has not been peer reviewed.
- All demonstration and test data is synthetic.
- Test counts, corpus distributions, and deterministic outcomes are software verification, not empirical validation.
- There is no evidence in this prototype that correspondence predicts revenue, purchase, switching, or conversion.
- Buyer conditions can become stale; Engine 2 relies on Engine 1 currentness status and does not independently decide BAEC validity.
- The Engine 1 → Engine 2 handoff is point in time and not continuously synchronized.
- No automatic outreach, CRM automation, account-state transition, or seller action exists.
- Reviewer identity in the research prototype is self-asserted.
- No claim of production readiness or production security is made.

Release state

Engine 2 v1.0 is frozen at the tagged release commit. The public Streamlit demonstration is a separate post-v1.0 presentation layer committed on top of that release. The public demo does not move or replace the v1.0 tag.

Live public demo: https://baec-engine2.streamlit.app

GitHub repository: https://github.com/bodhikaiparks/baec-trigger-intelligence

Appendix A. Reproducing the successful public demonstration

This appendix records the tested successful path through the public Engine 2 demonstration. Nothing is pre-filled; the visitor makes each human judgment. The expected final technical outcome is HUMAN_VERIFIED_CORRESPONDENCE.

Stages 1, 2, and 3

1. Open https://baec-engine2.streamlit.app and click Start Demo.
2. Step 1 - What the buyer said: read the confirmed buyer statement, threshold More than 8%, and timing July renewal. Click Continue to Monitoring.
3. Step 2 - Monitoring: review the Watching for sentence and the two authorized sources. Click See New Evidence.
4. Step 3 - New evidence: confirm Buyer threshold More than 8%, Current usable measurement 9.00%, and Relevant timing July renewal.
5. Note the correction sequence: 12.00% initially reported → corrected → 9.00% current. The 12.00% measurement remains in history but is invalidated.
6. Click Review Possible Correspondence.

Stage 4A - Supplier and event

Table A1. Human findings for Supplier and event.

Field	Entry	Evidence
Entity match	Supported	OBS-DEMO-02
Relationship match	Not applicable	
Source relevance	Supported	OBS-DEMO-02
Event or condition type	Supported	OBS-DEMO-02
Reason	The corrected renewal notice applies to this contract and comes from an authorized supplier notice source.	

Click Next.

Stage 4B - Buyer’s threshold

Table A2. Human findings for the buyer’s threshold.

Field	Entry	Evidence
Threshold match	Supported	DM-DEMO-02
Comparator match	Supported	DM-DEMO-02
Numeric magnitude	Supported	DM-DEMO-02
Unit match	Supported	DM-DEMO-02
Reason	The corrected contract price changes from USD 40,000 to USD 43,600, a 9.00% increase, which is more than the buyer’s 8% threshold.	

Click Next.

Stage 4C - Timing and context

Table A3. Human findings for timing and context.

Field	Entry	Evidence
Timing match	Not applicable	
Renewal or effective-date match	Supported	OBS-DEMO-02
Causal or contextual relevance	Not applicable	
Reason	The corrected notice takes effect on 1 July 2027, matching the July renewal context.	

Important: do not mark Timing match as Supported in this scenario. The Monitoring Plan marks Timing match Not applicable. A review that marks it Supported is refused by the locked domain rather than classified.

Click Next.

Stage 4D - Evidence quality and conclusion

Table A4. Final review entries for the successful path.

Field	Entry
Freshness	No concern
Contradictory evidence	No concern
Missing evidence	No concern
Provenance quality	No concern
Evidence sufficient?	Yes, sufficient
Revalidate buyer condition?	No
Reviewer alias	BP or another non-identifying alias

7. Click Submit Human Review.
8. Expected public headline: Verified correspondence.
9. Expected technical outcome badge: HUMAN_VERIFIED_CORRESPONDENCE.
10. Confirm the Why this matched summary shows More than 8%, 9.00%, and July renewal.
11. Read What this does NOT mean. No opportunity, outreach authorization, purchase intent, or account-state change is created.

The result is derived rather than hard-coded: all eight required dimensions are supported, the plan’s three non-required dimensions are marked Not applicable, the evidence is judged sufficient, and BAEC revalidation is not required.

Appendix B. Reproducing the other outcomes

The same public demo can reach the other locked outcomes through different human judgments. These examples demonstrate that the outcome is derived by the rule rather than selected by the interface.

Possible correspondence

Start from the successful path and make any one of the following changes:

- Set Is the evidence sufficient? to Cannot tell.
- Mark a required threshold dimension such as Threshold match or Unit match as Unresolved.
- Set BAEC revalidation to Required and provide a revalidation reason.

Expected result: POSSIBLE_CORRESPONDENCE. Correspondence is not verified because a required element remains unresolved, the evidence is not sufficient, or revalidation is required.

No correspondence

Start from the successful path and mark a required dimension contradicted. A clean example is Entity match = Contradicted.

Expected result: NO_CORRESPONDENCE. Under the locked rule, any contradicted required dimension yields NO_CORRESPONDENCE.

Domain refusal is not a fourth outcome

For this scenario, Timing match is explicitly not applicable. Marking Timing match Supported is refused by the locked domain. The visitor’s answers remain available for correction. A refused review is not NO, POSSIBLE, or HUMAN_VERIFIED correspondence; it is a validation boundary that prevents an invalid review from becoming authoritative.

Reference identifiers

Table B1. Public demonstration identifiers.

Reference	Identifier
Synthetic account	Alder Ridge Clinics
Synthetic supplier	Cobalt Imaging Supply
Corrected renewal notice	OBS-DEMO-02
Current usable measurement	DM-DEMO-02
Invalidated prior measurement	DM-DEMO-01
Refused forum item	ITEM-DEMO-FORUM-01
Release BAEC ID	BAEC-RC-001
Release Monitoring Plan	MP-RC-001 v1
Release candidate	CC-RC-001
Release review	REV-CC-RC-001-1

Project source reference

The public brief is a synthesis of the Engine 2 domain contract, release record, reproducibility record, release demonstration, public-demo implementation, and tests. Where this brief conflicts with a governing source or locked code, the governing source and locked code control.

Engine 2 source brief: https://github.com/bodhikaiparks/baec-trigger-intelligence/blob/main/docs/engine2/ENGINE2_TECHNICAL_RESEARCH_BRIEF.md

Public demo: https://baec-engine2.streamlit.app